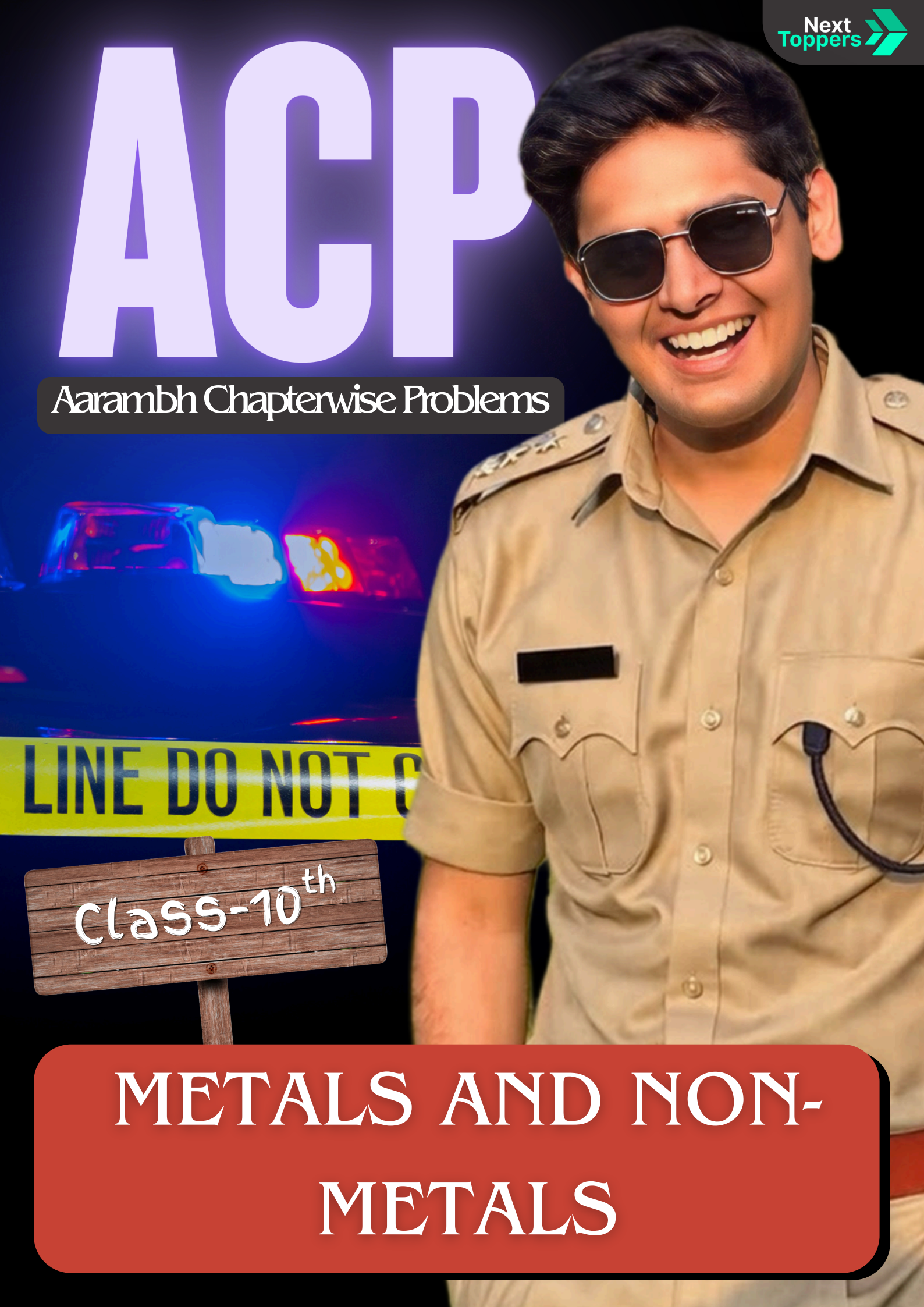


ACP

Aarambh Chapterwise Problems



LINE DO NOT CROSS

Class-10th

METALS AND NON-
METALS

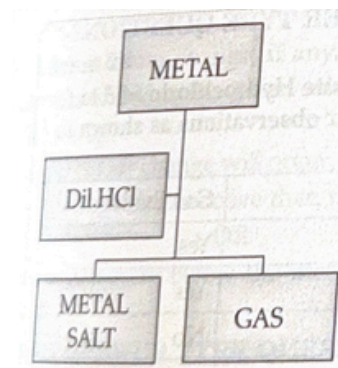
METALS AND NON-METALS

PART 1 (Basic MCQs)

Read the question properly!

1. Which of the following two combinations are correct?

	Metal	Gas Evolved
(i)	Copper	Yes
(ii)	Iron	Yes
(iii)	Magnesium	No
(iv)	Zinc	Yes



- (A) (i) and (iii)
- (B) (i) and (iv)
- (C) (ii) and (iii)
- (D) (ii) and (iv)

2. Three substances P, Q and R show the following properties:

Substance	Malleable	Conducts Electricity	Sonorous
P	Yes	Yes	Yes
Q	No	Yes	No
R	No	No	No

Which conclusion is most appropriate?

- (a) P and Q must both be metals
- (b) P is a metal; Q may be a non-metal such as graphite
- (c) R must be a metal
- (d) Conductivity alone is sufficient to classify an element

3. The electronic configurations of X, Y and Z are:

$$X = 2, 8, 1$$

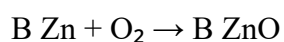
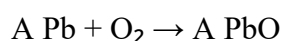
$$Y = 2, 8, 7$$

$$Z = 2, 8, 8$$

Which statement is correct?

- (a) X and Y are metals
- (b) X is a metal and Y is a non-metal
- (c) Y and Z form an ionic compound readily
- (d) Z readily gains one electron

4. Consider:



JOSH
METER



After balancing the equations, the values of A and B respectively are:

- (a) 1, 1
- (b) 2, 2
- (c) 2, 1
- (d) 4, 2

5. Equal-sized pieces of Mg, Al, Zn, Fe and Cu are placed separately in dilute HCl under identical conditions. Which observation is correct?

- (a) Cu reacts most vigorously
- (b) Fe does not react but Cu does
- (c) Mg reacts most vigorously while Cu shows no reaction
- (d) All evolve hydrogen at the same rate

6. Four substances show the following properties:

Substance	Melting Point	Conductivity in Solid State	Conductivity when Molten
A	Low	Good	Good
B	High	Poor	Good
C	High	Poor	Good
D	Low	Poor	Poor

Which are most likely ionic compounds?

- (a) A and B
- (b) B and C
- (c) C and D
- (d) A and D

7. A metal X is stored under kerosene and cannot be obtained from its oxide by reduction with carbon. X is most likely:

- (a) Cu
- (b) Fe
- (c) Na
- (d) Zn

8. An electrician needs an alloy that melts at a comparatively low temperature for joining electrical wires. Which is most suitable?

- (a) Brass
- (b) Bronze
- (c) Solder
- (d) Stainless steel

9. Consider the statements:

- I. Sodium reacts violently with cold water.
- II. Calcium reacts with cold water but generally does not catch fire.
- III. Iron reacts readily with cold water.
- IV. Copper does not react with water.

JOSH
METER



Correct statements are:

- (a) I and II only
- (b) I, II and IV
- (c) II, III and IV
- (d) All four

Questions 10 to 12 are Assertion–Reason based questions.

Each question consists of two statements: Assertion (A) and Reason (R). Select the correct answer from the options given below:

- (a) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
- (c) Assertion (A) is true, but Reason (R) is false.
- (d) Assertion (A) is false, but Reason (R) is true.

10. Assertion (A): Solid magnesium chloride does not conduct electricity.

Reason (R): Its ions are not free to move in the solid state.

11. Assertion (A): Aluminium generally shows much less corrosion than iron even though aluminium is more reactive.

Reason (R): Aluminium develops a thin, adherent oxide layer which prevents further attack.

12. Assertion (A): Sulphide and carbonate ores are usually converted into oxides before reduction.

Reason (R): Metal oxides are generally easier to reduce than the corresponding sulphides and carbonates.

PART 2 (Subjective Questions)

tho jaayenge aaram se!

VERY SHORT ANSWER

13. Copper is preferred over iron for making hot-water tanks. Give a reason and write the relevant reaction that could occur if hot iron were exposed to steam.

14. Hydrogen gas is generally not evolved when metals react with nitric acid. Explain why. Name the two metals which are exceptions when very dilute HNO_3 is used.

15. Aluminium is high in the reactivity series, yet aluminium utensils remain usable for years without severe corrosion. Explain.

16. Solid sodium chloride does not conduct electricity whereas molten sodium chloride does. Explain in terms of its particles.

17. Ionic compounds generally have high melting and boiling points. Explain the reason.

18. Why cannot carbon be used to extract sodium or magnesium from their oxides? State the method used instead.

19. Calcium starts floating when added to water, although calcium itself is denser than water. Explain. Why does the evolved gas usually not catch fire?

20. Pure metals are often converted into alloys. State any two desirable changes in properties produced by alloying and give the composition of solder.

COMPETENCY / APPLICATION QUESTIONS

21. A student records the following observations after adding dilute HCl:

Metal	Student's Observation
Copper	Hydrogen evolved
Iron	Hydrogen evolved
Magnesium	No reaction
Zinc	Hydrogen evolved

- Identify the two incorrect observations.
- Correct them.
- Write one balanced equation for a reaction that actually occurs.

22. A light-green solution of metal sulphate PSO_4 is taken in a beaker. A rod of metal Q is immersed in it. After some time, the green colour fades considerably and metal P gets deposited on Q.

If P is iron and Q is one of Zn, Cu or Ag:

- Identify Q.
- Write the balanced reaction.
- What does this experiment establish about the relative reactivities of P and Q?

23. Two white metal oxides X and Y are MgO and Al_2O_3 , but their labels have been removed.

- Suggest a chemical test using dilute HCl and NaOH to distinguish them.
- Identify the amphoteric oxide.
- Write one equation showing its amphoteric behaviour.

24. Ore P releases CO_2 when strongly heated in limited air. Ore Q releases SO_2 when strongly heated in excess air.

- Identify the type of ore P and Q.
- Name the metallurgical treatment for each.
- Write one balanced equation for either process.

25. A metal X has electronic configuration 2,8,1.

It is stored under kerosene, forms XCl on reaction with chlorine, and is obtained by electrolysis of its molten chloride.

- Identify X.
- Why cannot its oxide be reduced with carbon?
- Write the reaction occurring at the cathode during its extraction.

26. During electrolytic refining of copper, after some time:

- the impure copper electrode becomes thinner,
- the pure copper electrode becomes thicker,
- and a sludge collects below one electrode.

Explain all three observations and write the electrode reaction occurring at the cathode.

27. Compound X has the following properties:

- hard crystalline solid,
- high melting point,
- soluble in water,
- does not conduct electricity in solid state,
- conducts electricity when molten.

- (a) What type of compound is X?
(b) Explain the last two observations.

28. Aluminium has atomic number 13 and oxygen has atomic number 8.

- (a) Show how aluminium and oxygen form Al_2O_3 by transfer of electrons.
(b) Name the cation and anion formed.
(c) Why is the resulting compound electrically neutral?

29. Three metals X, Y and Z are tested with their salt solutions:

Metal added	XSO_4	YSO_4	ZSO_4
X	—	Reaction	Reaction
Y	No reaction	—	Reaction
Z	No reaction	No reaction	—

- (a) Arrange X, Y and Z in decreasing order of reactivity.
(b) Which metal is easiest to displace from its salt solution?
(c) Explain your reasoning.

30. Pure gold is 24-carat and is generally not preferred for ordinary jewellery.

- (a) Why?
(b) What is meant by 22-carat gold?
(c) Why does alloying make gold more suitable for ornaments?

31. Metal X can displace iron from FeSO_4 solution but cannot displace zinc from ZnSO_4 .

Predict whether X will react with:

- (i) CuSO_4
(ii) AgNO_3

Also state where X lies relative to Zn and Fe in the reactivity series.

32. An element A forms two oxides, AO and AO_2 . AO is neutral whereas AO_2 is acidic.

- (a) Is A more likely to be a metal or a non-metal?
(b) Give a possible example of A.
(c) Explain your conclusion from the nature of its oxides.

PART 3 (Advanced Questions)

LONG ANSWER / HIGHER-ORDER

33. Element X has electronic configuration 2, 8, 2 and element Y has configuration 2, 8, 7.

- (a) Identify whether X and Y are metals or non-metals.
(b) Show the formation of the compound between X and Y by electron transfer.

- (c) Write its formula.
- (d) State two physical properties of the compound formed.
- (e) Explain why it conducts electricity in molten state but not in solid state.

34. Four metals P, Q, R and S give the following results:

- P displaces Q and R from their salt solutions.
- Q displaces R but not P.
- S cannot displace any of the other three.
- R cannot displace Q.

- (a) Arrange them in decreasing order of reactivity.
- (b) Which metal is most likely to occur in the free state?
- (c) Predict whether P will react with S-salt solution.
- (d) Write a general equation representing this reaction.
- (e) Explain how displacement reactions help construct the reactivity series.

35. Mercury, zinc and sodium lie in different regions of the reactivity series. Compare their extraction under the following heads:

- (a) position in the reactivity series,
- (b) general form in which they occur,
- (c) treatment of the ore/compound,
- (d) method by which the metal is finally obtained,
- (e) one relevant balanced equation for each metal.

36. An engineer has to select suitable metals/alloys for:

1. electrical wiring,
2. hot-water tank,
3. jewellery,
4. joining electrical wires,
5. an object requiring high corrosion resistance.

For each application:

- (a) suggest a suitable metal/alloy, and
- (b) justify the choice using a relevant physical or chemical property.

37. A manufacturer has pure iron, copper and 24-carat gold but decides to use stainless steel, brass and 22-carat gold instead. Answer:

- (a) Why is stainless steel preferred to pure iron in many applications?
- (b) State the major metals present in stainless steel.
- (c) What are the constituents of brass?
- (d) Why is 24-carat gold alloyed before making jewellery?
- (e) State one general change in electrical conductivity caused by alloying.

38. A laboratory provides the following metals: Mg, Zn, Fe, Cu and Au

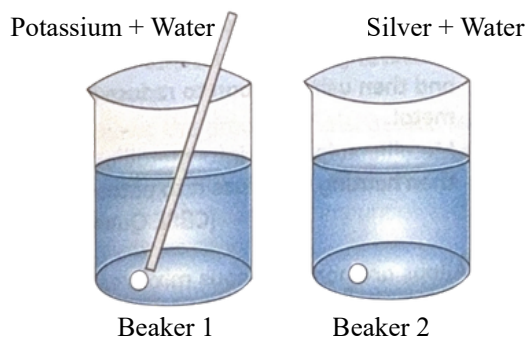
and the following reagents: dilute HCl, concentrated HNO_3 and freshly prepared aqua regia.

Design an investigation to answer:

- (A) Which metal would react most readily with dilute HCl?
- (B) Which of the given metals would not displace hydrogen from dilute HCl?
- (C) Why should the result with nitric acid not be used blindly to place metals relative to hydrogen?
- (D) Which reagent can dissolve gold when the individual acids cannot? State its composition.
- (E) Explain how the observations collectively show that different metals have different chemical reactivities.

CASE-BASED QUESTIONS

39. A student adds a small piece of potassium to water in Beaker A and a small piece of silver to water in Beaker B, as shown in the figure.



Observe the figure carefully and answer the following questions:

- In which beaker will a chemical reaction occur?
- Name the gas evolved during the reaction.
- Write the balanced chemical equation for the reaction.
- Explain why potassium reacts vigorously with water whereas silver does not react with water, using their positions in the reactivity series.

40. Study the given reactivity series carefully and answer the following questions:

- What happens when a zinc strip is placed in copper sulphate solution? Write the observation.
- Why does copper fail to displace zinc from zinc sulphate solution?
- Among Mg, Zn, Cu and Ag, which metal can be displaced from its salt solution by all the other three metals?
- Write one balanced chemical equation for a displacement reaction that supports your answer in part (c).

Most reactive	
Potassium	K
Sodium	Na
Calcium	Ca
Magnesium	Mg
Aluminium	Al
Carbon	C
Zinc	Zn
Iron	Fe
Tin	Sn
Lead	Pb
Hydrogen	H
Copper	Cu
Silver	Ag
Gold	Au
Platinum	Pt
Least reactive	

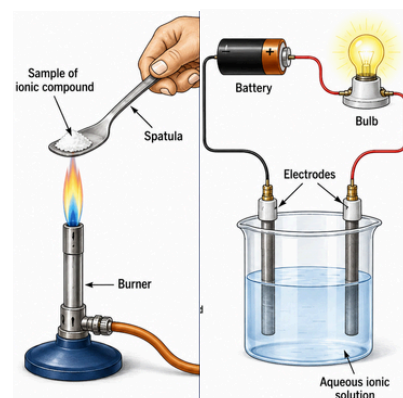
41. Many pure metals are too soft or possess unsuitable properties for direct use, so alloys are commonly used.

- What is an alloy?
- How does alloying generally affect electrical conductivity?
- Name the alloy used for joining electrical wires and give its constituents.
- Why is stainless steel preferred over pure iron for many utensils?

42. Study the given activity showing the properties of ionic compounds and answer the following questions:

Different samples of ionic compounds are heated, tested for solubility in different solvents, and their aqueous solutions are connected in an electric circuit as shown in the figure.

- What does the heating experiment indicate about the melting points of ionic compounds?
- In which type of solvent are ionic compounds generally soluble?
- Will the bulb glow when an aqueous solution of an ionic compound is used in the circuit? Give a reason.
- State any two characteristic properties of ionic compounds that can be inferred from the above activity.



43. Study the given data showing the change in mass of two iron objects before and after rusting and answer the following questions:

Student	Object	Mass of Object before Rusting (g)	Mass of Object after Rusting (g)
A	Nail	3	3.15
B	Thin plate	6	6.33

A student takes an iron nail and a thin iron plate and exposes them to conditions suitable for rusting. After some time, an increase in their masses is observed.

- Why does the mass of an iron object increase after rusting?
- Two iron objects of equal mass but different shapes may show different amounts of rusting under identical conditions. Give a reason.
- What is galvanisation?
- A galvanised iron article remains protected from rusting even when the zinc coating is scratched. Explain why.

44. Old copper domes gradually become green on exposure to the atmosphere.

- Name the green coating formed.
- Why should replacing the copper dome with an ordinary iron dome not be considered an improvement?
- Which property allows copper to be shaped into thin curved sheets?
- Does the green coating necessarily make copper unsuitable for long-term outdoor use? Explain.

45. During copper refining, the following masses are measured:

Electrode	Initial mass	Final mass
Electrode P	250 g	223 g
Electrode Q	100 g	125 g

A small quantity of insoluble material is also found below P.

- Which electrode is the anode?
- Which is the cathode?
- Why does P lose mass while Q gains mass?
- What is the material collected below P called?

46. A metallurgist receives three ores:

- Ore A: an oxide of a very low-reactivity metal
- Ore B: a carbonate of a moderately reactive metal
- Ore C: a chloride of a highly reactive metal

For each ore:

- State the principal extraction approach.
- Which ore requires calcination before reduction?
- Which requires electrolysis?
- Explain why the same extraction method cannot be used for all three.

- Physical Properties of Metals and Non-metals & Their Uses – Q2, Q36, Q44(C)
- Reaction with Oxygen & Nature of Oxides – Q4, Q11, Q15, Q23, Q32
- Reaction of Metals with Water – Q9, Q13, Q19, Q39
- Reaction of Metals with Acids – Q1, Q5, Q14, Q21, Q38
- Reaction of Metals with Salt Solutions & Reactivity Series – Q22, Q29, Q31, Q34, Q40
- Reaction of Metals with Non-metals & Properties of Ionic Compounds – Q3, Q6, Q10, Q16, Q17, Q27, Q28, Q33, Q42
- Occurrence and Extraction of Metals – Q7, Q12, Q18, Q24, Q25, Q34(B), Q35, Q46
- Refining of Metals – Q26, Q45
- Corrosion – Q11, Q15, Q43, Q44
- Prevention of Corrosion & Alloys – Q8, Q20, Q30, Q36, Q37, Q41, Q43(C–D)



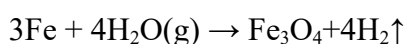
SOLUTIONS

MULTIPLE CHOICE QUESTIONS [MCQ's]

1. (d)
2. (b)
3. (b)
4. (b)
5. (c)
6. (b)
7. (c)
8. (c)
9. (b)
10. (a)
11. (a)
12. (a)

VERY SHORT ANSWERS

13. Copper is used for hot-water tanks because it does not react with hot water or steam, whereas iron reacts with steam.



14. Nitric acid is a strong oxidising agent. It oxidises the hydrogen initially produced to water and itself gets reduced to nitrogen oxides. Hence, H_2 is generally not evolved.

Exceptions: Magnesium and manganese react with very dilute HNO_3 to evolve hydrogen.

15. Although aluminium is highly reactive, it develops a thin, tough and protective Al_2O_3 layer on its surface. This layer prevents air and moisture from coming in contact with the underlying aluminium and prevents further corrosion.

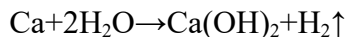
16. In solid NaCl , Na^+ and Cl^- ions are fixed in the crystal lattice and cannot move.

In molten NaCl , the ions become free to move and therefore conduct electricity.

17. Ionic compounds contain oppositely charged ions held together by strong electrostatic forces of attraction. A large amount of heat energy is required to overcome these forces; hence they have high melting and boiling points.

18. Na and Mg are more reactive than carbon and have a greater affinity for oxygen. Therefore, their oxides cannot be reduced by carbon. They are extracted by electrolytic reduction of their molten compounds.

19. Calcium reacts with water:



Bubbles of hydrogen stick to the surface of calcium, making it float. The reaction is not sufficiently exothermic to ignite the hydrogen produced.

20. Two desirable effects of alloying are:

- increases hardness/strength,
- increases resistance to corrosion.

Solder = Lead + Tin (Pb + Sn).

COMPETENCY / APPLICATION QUESTIONS

21. Incorrect observations:

- Copper – Hydrogen evolved ✗
- Correct: Copper shows no reaction with dilute HCl.
- Magnesium – No reaction ✗
- Correct: Magnesium reacts vigorously and evolves hydrogen.

For example: $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\uparrow$

Iron and zinc observations are correct.

22. Since P is Fe and metal P gets deposited on Q, Q must be more reactive than iron.

Among Zn, Cu and Ag: $\text{Q} = \text{Zn}$

Reaction: $\text{Zn} + \text{FeSO}_4 \rightarrow \text{ZnSO}_4 + \text{Fe}$

Hence: $\text{Zn} > \text{Fe}$

23. X and Y are MgO and Al_2O_3 .

Both react with HCl, but on treatment with NaOH:

- Al_2O_3 reacts
- MgO does not react

Therefore Al_2O_3 is amphoteric.

$\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$

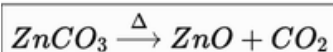
It also reacts with acids: $\text{Al}_2\text{O}_3 + 6\text{HCl} \rightarrow 2\text{AlCl}_3 + 3\text{H}_2\text{O}$

24. Ore P: carbonate ore

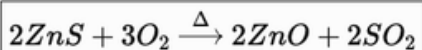
Ore Q: sulphide ore

P undergoes calcination.

Example:



Q undergoes roasting.



25. (a) X = Sodium (Na)

(b) Sodium is more reactive than carbon and its oxide is very stable; hence it cannot be reduced by carbon.

(c) Cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$

26. In electrolytic refining of copper:

Anode: impure copper

Cathode: pure copper

At anode: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^{-}$

Therefore, the impure copper anode becomes thinner.

At cathode: $\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$

Pure copper gets deposited, so the cathode becomes thicker.

Insoluble impurities settle below the anode as anode mud.

27. X is an ionic compound.

In solid state, ions are fixed and cannot move, so it does not conduct electricity.

When molten, ions become mobile and carry electric charge, so it conducts electricity.

28. Al = 2, 8, 3

O = 2, 6

Each Al loses three electrons: $\text{Al} \rightarrow \text{Al}^{3+} + 3\text{e}^{-}$

Each O gains two electrons: $\text{O} + 2\text{e}^{-} \rightarrow \text{O}^{2-}$

For complete electron transfer: $2\text{Al} \rightarrow 2\text{Al}^{3+} + 6\text{e}^{-}$

$3\text{O} + 6\text{e}^{-} \rightarrow 3\text{O}^{2-}$

Hence: Al_2O_3

Cation: Al^{3+}

Anion: O^{2-}

Total positive charge: $2(+3) = +6$

Total negative charge: $3(-2) = -6$

Hence the compound is electrically neutral.

29. From the table:

X displaces Y and Z: $\text{X} > \text{Y}, \text{X} > \text{Z}$

Y displaces Z: $\text{Y} > \text{Z}$

Therefore: $\text{X} > \text{Y} > \text{Z}$

(b) Z is easiest to displace because it is the least reactive.

30. (a) 24-carat gold is very soft and can easily get deformed, so it is unsuitable for ordinary jewellery.

(b) 22-carat gold means 22 parts pure gold + 2 parts another metal, usually Cu or Ag, out of 24 parts.

(c) Alloying increases the hardness and strength of gold.

31. X displaces Fe: $\text{X} > \text{Fe}$

But cannot displace Zn: $\text{Zn} > \text{X}$

Therefore: $\text{Zn} > \text{X} > \text{Fe}$

Since Cu and Ag lie below Fe:

(i) X will displace Cu from CuSO_4 .

(ii) X will displace Ag from AgNO_3 .

So X is less reactive than Zn but more reactive than Fe, Cu and Ag.

32. A possible element is carbon.

CO = neutral oxide

CO₂ = acidic oxide

Therefore:

- (a) A is a non-metal.
 (b) A can be carbon.
 (c) Non-metals generally form acidic or neutral oxides.

LONG ANSWER QUESTIONS

33. $X = 2, 8, 2 \rightarrow$ Mg, metal

$Y = 2, 8, 7 \rightarrow$ Cl, non-metal

Mg loses two electrons: $Mg \rightarrow Mg^{2+} + 2e^{-}$

Each Cl gains one: $2Cl + 2e^{-} \rightarrow 2Cl^{-}$

Hence: $MgCl_2$

Two properties:

- high melting and boiling points,
- generally soluble in water,
- hard and brittle,
- conduct electricity when molten/aqueous.

In solid state ions are fixed; in molten state they are free to move

34. Given:

- P displaces Q and $R \rightarrow P > Q, R$
- Q displaces R $\rightarrow Q > R$
- S cannot displace any $\rightarrow$ S is least reactive

Therefore: $P > Q > R > S$

(b) S is most likely to occur in free state because it is least reactive.

(c) Yes, P will displace S.

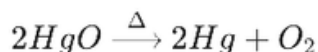
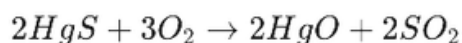
General reaction: $P + \text{salt of } S \rightarrow \text{salt of } P + S$

Displacement experiments allow us to compare relative reactivities: a metal that displaces another is more reactive.

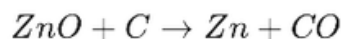
35.

Metal	Position	Common occurrence	Extraction
Hg	Low	HgS (cinnabar)	Heating/roasting followed by decomposition
Zn	Middle	ZnS/ZnCO ₃	Roasting/calcination $\rightarrow$ oxide $\rightarrow$ carbon reduction
Na	Top	NaCl	Electrolysis of molten salt

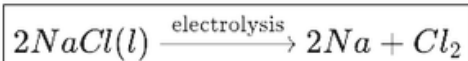
Mercury:



Zinc:



Sodium:



36.

Application	Suitable Material	Reason
Electrical wiring	Copper	Excellent conductor + ductile
Hot-water tank	Copper	Does not react with hot water/steam
Jewellery	Gold	Lustrous, malleable and unreactive
Joining wires	Solder	Low melting point
High corrosion resistance	Stainless steel	Strong and corrosion-resistant

37.(a) Stainless steel is stronger and much more corrosion-resistant than pure iron.

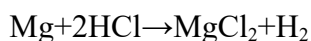
(b) Major metals: Iron, chromium and nickel.

(c) Brass: Cu+Zn

(d) 24-carat gold is too soft; alloying increases its hardness.

(e) Alloying generally reduces electrical conductivity compared with the pure metal.

38.(A) Mg would react most readily with dilute HCl.



(B) Both Cu and Au will not displace hydrogen from dilute HCl.

(C) HNO_3 is a strong oxidising agent. It generally oxidises H_2 to water, so absence of hydrogen does not necessarily mean the metal lies below hydrogen.

(D) Aqua regia dissolves gold.

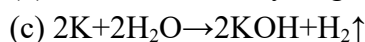
Conc. HCl : Conc. HNO_3 = 3:1

(E) The different reactions show that metals differ in their tendency to lose electrons and hence have different chemical reactivities.

CASE-BASED QUESTIONS

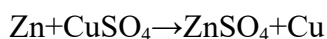
39.(a) Reaction occurs in Beaker A (potassium + water).

(b) Gas evolved = Hydrogen



(d) Potassium is near the top of the reactivity series and reacts vigorously with water. Silver lies near the bottom and does not react with water.

40.a) Zinc displaces copper:

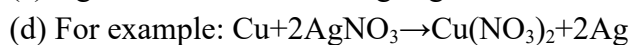


Observation:

- blue colour of CuSO_4 fades,
- reddish-brown copper deposits on zinc.

(b) Copper is less reactive than zinc, so it cannot displace zinc from ZnSO_4 .

(c) Ag is least reactive among Mg, Zn, Cu and Ag and can therefore be displaced by the other three.



41.(A) An alloy is a homogeneous mixture of two or more metals, or a metal with a non-metal.

(B) Alloying generally decreases electrical conductivity.

(C) Solder = lead + tin (Pb + Sn).

(D) Stainless steel is harder, stronger and more resistant to corrosion than pure iron, so it is suitable for utensils.

42. (a) Ionic compounds generally have high melting points and do not melt easily on ordinary heating because of strong electrostatic forces between ions.

(b) They are generally soluble in polar solvents such as water and insoluble in kerosene/petrol.

(c) Yes, the bulb will glow.

In aqueous solution, ions are free to move and carry electric current.

(d) Any two:

- high melting and boiling points,
- generally soluble in water,
- insoluble in organic solvents,
- conduct electricity in molten/aqueous state.

43.(a) Iron combines with oxygen and moisture during rusting, adding oxygen/water to the iron and forming hydrated iron(III) oxide. Hence mass increases.

Rust: $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$

(b) Different shapes have different surface areas exposed to air and moisture. Greater exposed surface area $\rightarrow$ greater rusting.

(c) Galvanisation is coating iron/steel with a thin layer of zinc to protect it from rusting.

(d) Zinc is more reactive than iron and undergoes oxidation preferentially. Thus it provides sacrificial protection even if the coating is scratched.

44.(A) The green coating is basic copper carbonate: $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$

(B) Iron would rust, and rust is porous/flaky and does not provide a protective layer. The copper patina protects the underlying copper.

(C) Malleability

(D) No. The green patina can act as a protective barrier and slow further corrosion of copper.

45. P loses mass, therefore:

(A) P = Anode, Q gains mass:

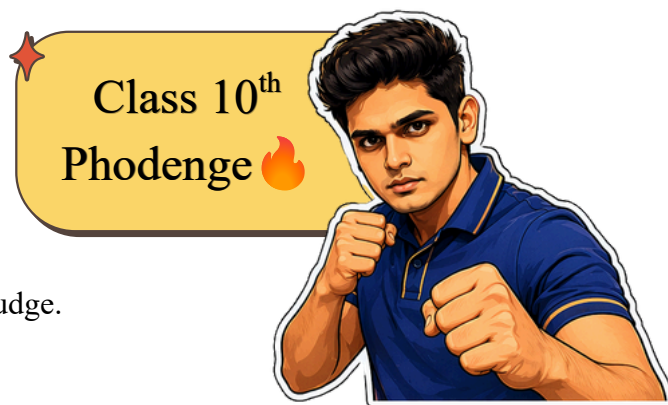
(B) Q = Cathode

(C) At P: $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$

Copper dissolves into solution.

At Q: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ (Copper gets deposited.)

(D) Insoluble material below P is called anode mud / anode sludge.



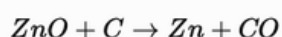
46. Ore A: oxide of a very low-reactivity metal

$\rightarrow$ metal can generally be obtained by heating/decomposition of the oxide.

Ore B: carbonate of a moderately reactive metal

$\rightarrow$ first calcination, then reduction of oxide.

Example: $\text{ZnCO}_3 \xrightarrow{\Delta} \text{ZnO} + \text{CO}_2$



Ore C: chloride of a highly reactive metal $\rightarrow$ electrolysis of molten chloride.

Therefore:

(B) Ore B requires calcination.

(C) Ore C requires electrolysis.

(D) The same extraction method cannot be used because the stability of metal compounds and ease of reduction depend on the reactivity of the metal. Highly reactive metals cannot be reduced by carbon, while less reactive metals can be obtained by much simpler methods